

Switch and Router Configuration BasicsTheory

1) Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Step 1: Open Cisco Packet Tracer

- Open Cisco Packet Tracer and create a new project.
- Add one switch to the workspace.

Step 2: Open the Switch CLI

- Click the switch and open the CLI tab.
- Enter privileged EXEC mode and then global configuration mode.

Step 3: Create VLAN 10

- Create VLAN 10.
- Name VLAN 10 as Sales.

Step 4: Create VLAN 20

- Create VLAN 20.
- Name VLAN 20 as Support.

Verification

- Use the show vlan brief command to verify that VLAN 10 (Sales) and VLAN 20 (Support) exist.

Screenshot to Include

- Switch CLI showing VLAN 10 Sales and VLAN 20 Support.

2) Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Step 1: Assign Ports to VLAN 10

- Select three switch ports using the interface range command.
- Configure the selected ports as access ports.
- Assign the ports to VLAN 10 (Sales).

Step 2: Assign Ports to VLAN 20

- Select two different switch ports using the interface range command.
- Configure the selected ports as access ports.
- Assign the ports to VLAN 20 (Support).

Step 3: Verify VLAN Membership

- Run show vlan brief.
- Confirm that the three selected ports are listed under VLAN 10 (Sales).
- Confirm that the two selected ports are listed under VLAN 20 (Support).

Example Port Assignment

- VLAN 10 (Sales): FastEthernet0/1, FastEthernet0/2, FastEthernet0/3.
- VLAN 20 (Support): FastEthernet0/4, FastEthernet0/5.

Screenshot to Include

- CLI output of show vlan brief showing the correct port membership.

3) Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Step 1: Add and Connect the Routers

- Add Router 1 and Router 2 to the simulation.
- Connect the routers using a serial or Ethernet link.

Step 2: Configure Router Interfaces

- Assign an IP address to the interface on Router 1 connected to Router 2.
- Assign an IP address to the corresponding interface on Router 2.
- Use addresses from the same subnet on the router-to-router link.
- Make sure the interfaces are enabled.

Example Addressing

- Router 1 link interface: 10.0.0.1/30.
- Router 2 link interface: 10.0.0.2/30.
- LAN behind Router 2: 192.168.20.0/24.

Step 3: Configure a Static Route on Router 1

- On Router 1, create a static route to the LAN behind Router 2.
- Use Router 2's link IP address as the next hop.
- The destination network in this example is 192.168.20.0/24.

Step 4: Verify the Route

- Use show ip route on Router 1.
- Confirm that the static route appears in the routing table.
- Use ping or another connectivity test to verify reachability.

Screenshot to Include

- Router 1 routing table showing the static route.
- Router interface configuration showing the assigned IP addresses.

4) Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Step 1: Connect Devices to the VLANs

- Connect a device to a port assigned to VLAN 10 (Sales).
- Connect another device to a port assigned to VLAN 20 (Support).
- Assign IP addresses from different subnets to the devices.

Step 2: Test Connectivity

- From the Sales device, ping the IP address of the Support device.
- Observe that the ping fails when no inter-VLAN routing is configured.

Why the Ping Fails

- VLAN 10 and VLAN 20 are separate Layer 2 broadcast domains.
- A Layer 2 switch does not route traffic between different VLANs.
- The devices therefore need a Layer 3 routing function to communicate between the VLANs.

Configuration That Allows Inter-VLAN Communication

- Use router-on-a-stick with a router and an 802.1Q trunk link to the switch.
- Alternatively, use a Layer 3 switch and configure SVIs for VLAN 10 and VLAN 20.
- Assign the appropriate default gateway to devices in each VLAN.

Verification

- After configuring inter-VLAN routing, repeat the ping.
- A successful reply confirms that routing between Sales and Support is working.

Screenshot to Include

- Failed ping before inter-VLAN routing.
- Successful ping after inter-VLAN routing is configured.